First year Aerospace Engineering

Waves and Vibrations exercises

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1) A simple pendulum takes 2.0 seconds to swing from its leftmost position to its rightmost position. What is the frequency of this pendulum in Hertz?

2) You are looking at a displacement-time graph for a single particle in a wave, and a displacement-position graph for the entire wave at one instant. On the displacement-time graph, the distance between two adjacent peaks is 0.5 seconds. On the displacement-position graph, the distance between a peak and the adjacent trough is 2.0 meters. Calculate the speed of this wave.

3) A hiker shouts toward a vertical cliff face 680 meters away. If the speed of sound in air is 340 m/s, how much time passes between the moment the hiker shouts and the moment they hear the echo?

4) Two astronauts are on a spacewalk outside the ISS, floating only 1 meter apart. Their radios fail. One astronaut bangs their wrench hard against the metal hull of the space station. Can the other astronaut hear the sound? Explain why or why not, distinguishing between the vacuum of space and the station's hull.

5) A wave generator in a ripple tank is set to a frequency of 10 Hz, producing waves with a speed of 0.5 m/s. If the frequency of the generator is doubled to 20 Hz, but the depth of the water (which determines wave speed) remains unchanged, what happens to the wavelength?

6) You see a flash of lightning and hear the thunder exactly 5 seconds later. If the speed of light is 3 * 10^8 m/s and the speed of sound is 340 m/s, calculate the distance to the lightning strike. (Do you need to account for the travel time of the light?).

7) Two particles in a transverse wave are separated by a horizontal distance of exactly half a wavelength 0.5 * λ. If the first particle is currently at its maximum positive displacement (a crest), describe the position and motion of the second particle.

8) You have a long slinky stretched out. To create a longitudinal wave, do you move your hand back and forth parallel to the slinky or side-to-side perpendicular to it? And in which of these two wave types does the medium (the slinky coils) travel from one end to the other?

9) A beam of red light (wavelength ~700 nm) and a beam of blue light (wavelength ~450 nm) are both traveling through a vacuum. Which beam travels faster?

10) A wave has a "peak-to-peak" amplitude of 10 meters. What is the mathematical amplitude A of this wave used in physics equations?

11) If 50 complete waves pass a fixed point in 5 seconds, what is the Period (T) of the wave?

12) An ocean wave travels East at 5 m/s. A specific water molecule is currently on the crest of the wave. After one full period, what is the net horizontal displacement of that water molecule?

13) Sound travels through steel at approximately 5960 m/s and through air at 343 m/s. If a sound wave with a fixed frequency of 1000 Hz travels from air into a steel rail, does its wavelength increase, decrease, or stay the same?

14) Two waves of the same frequency are traveling in the same direction. Wave A starts at t=0 with zero displacement and moving up. Wave B starts at t=0 with maximum positive displacement. What is the phase difference between these two waves?

15) Radio waves are electromagnetic waves. If a radio station broadcasts at 100 MHz, and radio waves travel at the speed of light, what is the length of one radio wave?

16) A guitar string is tightened, which increases the speed of waves traveling along the string. If the length of the string (which defines the wavelength of the fundamental note) stays the same, does the frequency of the sound produced increase or decrease?

17) In a longitudinal sound wave, the distance between the center of a compression and the center of the adjacent rarefaction is 1.5 meters. What is the wavelength of this sound wave?

18) In a transverse wave traveling to the right, particle P is currently passing through the equilibrium position moving upwards. Where will particle P be after one-quarter of a period?

19) Which of the following properties determines the pitch of a sound wave as heard by a human ear: Amplitude, Speed, or Frequency?

20) A duck is floating on a lake. It bobs up and down 15 times in one minute. The distance between wave crests is 4 meters. Calculate the wave speed.

21) Two radio towers are separated by a distance of 100 meters and are transmitting identical radio waves in phase with a wavelength of 20 meters. A car is driving on a straight road that runs parallel to the line connecting the towers, located 200 meters away from the towers. As the car drives along this road, will the driver detect a steady signal strength, or will the signal fade in and out? Explain in terms of path difference.

22) Two small speakers are placed 3.0 meters apart and are playing a pure tone of 170 Hz in phase. The speed of sound is 340 m/s. A microphone is placed exactly 4.0 meters in front of one of the speakers (forming a right triangle with the two speakers). Determine if the microphone is at a location of constructive interference (loud) or destructive interference (quiet).

23) Two wave sources, A and B, are separated by a distance. They emit identical waves with wavelength "lambda", but Source B is "out of phase" with Source A by 180 degrees (pi radians). If you stand at a point exactly equidistant from both sources (the perpendicular bisector), will you observe constructive or destructive interference? Explain why this differs from the standard "in-phase" case.

24) Two speakers are separated by a distance of 2.0 meters. An observer stands 3.0 meters directly in front of one of the speakers. The speakers are vibrating in phase. What is the lowest possible frequency (fundamental) for which the observer will hear a minimum (quiet) sound due to destructive interference? Assume the speed of sound is 343 m/s.

25) Light from a single laser passes through two narrow slits (Double Slit Experiment) and hits a screen. At a specific point P on the screen, the path difference between the light rays from the two slits is exactly 2.25 wavelengths. Is this point a bright fringe (maxima), a dark fringe (minima), or an intermediate intensity? Explain your reasoning.

26) A fighter jet is flying low over a calm ocean while communicating with an aircraft carrier. The radio antenna on the jet receives two signals: one that travels directly from the carrier and one that reflects off the surface of the water. The water acts as a "hard" boundary for radio waves, causing a 180-degree phase shift upon reflection. If the pilot flies at an altitude where the path difference between the direct and reflected ray is exactly one full wavelength, will the signal be strong (constructive) or weak (destructive)? Explain the "trap" involving the reflection.

27) To reduce the radar signature of a stealth aircraft, engineers apply a special coating to the fuselage. The goal is to prevent radar waves from reflecting back to the enemy station. This is done by ensuring the wave reflecting off the top surface of the coating destructively interferes with the wave reflecting off the metal skin underneath. What is the minimum thickness of this coating in terms of the radar wavelength (lambda) to achieve cancellation? (Assume the radar wave travels slower in the coating than in air, and ignore phase shifts at the boundaries for this specific simplification).

28) A propeller plane has two engines mounted on its wings, separated by a distance of 6.0 meters. Both propellers are spinning at a frequency that produces a 100 Hz hum, and they are perfectly synchronized (in phase). A noise sensor is placed on the fuselage, exactly halfway between the two engines. If the speed of sound is 340 m/s, is the noise at the sensor loud or quiet? If the pilot then shuts down one engine, does the volume at the sensor drop by half, or does it change more dramatically?

29) A modern spacecraft uses a "phased array" antenna, which has multiple small transmitters in a row, rather than a moving dish. To beam a signal directly forward (perpendicular to the array), all transmitters fire at the same time. To steer the beam to the right using interference, should the transmitters on the left side of the array fire earlier, later, or at the same time as the transmitters on the right? Explain in terms of wavefronts.

30) An air conditioning duct in a passenger jet contains a noise-cancellation system. A microphone picks up the "drone" of the fan, and a speaker further down the duct plays an "anti-noise" signal to cancel it out. If the fan noise has a wavelength of 2.0 meters, and the system is designed to create destructive interference purely by geometry (a passive delay loop), how much longer must the "bypass" path be compared to the direct path to silence the noise?

31) A mass M is suspended from the ceiling by two identical vertical springs, connected in series (one after the other), each with a stiffness constant of k. To stabilize the motion, a third identical vertical spring (stiffness k) connects the bottom of the mass directly to the floor. The mass is constrained to move only vertically. Construct the differential equation of motion for the vertical displacement x, and determine the natural frequency of this system.

32) A simple pendulum consisting of a bob of mass m and a massless string of length L is hanging inside an elevator car. The elevator is not stationary; it is accelerating upward with a constant acceleration of a = g/3. Construct the equation of motion for small angular oscillations (theta) and find the natural frequency of the pendulum relative to the elevator car.

33) A 5 kg block is attached to a horizontal spring with a stiffness of k = 500 N/m. The block slides on a surface that is covered in a thick fluid. This fluid exerts a drag force that is proportional to the velocity of the block, with a drag magnitude of 20 Newtons when the block moves at 1 m/s. Construct the full differential equation of motion (with numerical coefficients) and calculate the undamped natural frequency ($omega_n$).

34) A solid cylinder of mass M and radius R rolls without slipping on a horizontal surface. It is attached to a wall by a horizontal spring of stiffness k connected to its center of mass. Construct the equation of motion for the horizontal displacement x of the center of mass. What is the natural frequency?

35) A heavy door with a moment of inertia I is equipped with a torsional spring (stiffness K_t) and a rotational damper (damping coefficient c). You are told the system is "critically damped," meaning it returns to equilibrium as fast as possible without overshooting. Write the symbolic equation of motion for the angle theta, and derive the expression for the specific damping coefficient c required to achieve this state in terms of K_t and I.

36) On the International Space Station, weightless astronauts measure their mass using a Body Mass Measurement Device (BMMD), which is essentially a spring-mounted chair. When the chair is empty (mass = 10 kg), it oscillates with a period of 1.0 second. When an astronaut sits in the chair, the period increases to 2.5 seconds. Calculate the mass of the astronaut.

37) A navigation sensor (mass m) is suspended inside a rocket by a vertical spring (stiffness k). The rocket is sitting on the launchpad (1g). Later, the rocket blasts off and achieves a sustained vertical acceleration of 5g.
a) Does the natural frequency of the sensor's oscillation increase, decrease, or stay the same during the 5g burn compared to the 1g pre-launch state?
b) By what factor does the equilibrium position of the sensor shift (the "sag") during the 5g burn compared to the 1g pre-launch sag?

38) A simplified model of an aircraft wing behaves like a mass on a spring. The "spring" is the bending stiffness of the wing spar, and the "mass" is a fuel tank located at the wingtip. When the fuel tank is completely full, the wingtip oscillates with a frequency of 2.0 Hz. After a long flight, the tank is empty, and the total mass of the tip tank has decreased by a factor of 4 (i.e., Mass_empty = 0.25 * Mass_full). What is the new vibration frequency of the wingtip?

39) A drone with a mass of 5 kg is equipped with a vertical landing gear spring (k = 2000 N/m). The drone loses power and falls vertically from a height of 2 meters above the ground. It hits the ground and the spring compresses. To find the maximum compression of the spring, a student writes the equation: Potential Energy (mgh) = Spring Energy (0.5 * k * x^2). What specific energy term did the student forget to include that makes this calculation wrong for a vertical impact?

40) A mechanical accelerometer consists of a small proof mass attached to a spring inside a casing. The casing is bolted to the frame of a jet fighter. When the jet pulls a steady maneuver, the spring inside the accelerometer compresses by 2 mm. If the natural frequency of the accelerometer's internal mass-spring system is 100 Hz (approx 628 rad/s), calculate the acceleration the jet is undergoing in m/s^2. (Hint: The spring force must provide the acceleration).

41) A satellite component of mass m is hung vertically from a spring of stiffness k in a lab on Earth. The component hangs motionless, stretching the spring by a distance "d" (where d = mg/k). The system is then locked in this exact position and launched into deep space (zero gravity). Once in space, the locking mechanism is released. Describe the resulting motion of the component: What is the frequency, and specifically, what is the amplitude of the oscillation?

42) An engine mount is modeled as a mass-spring system with a natural frequency of 40 Hz. The engine rotates at 2400 RPM (Revolutions Per Minute). Is this a safe design? Explain why or why not by converting the engine speed to Hz and comparing it to the natural frequency.

43) Two satellites, Mass A (1000 kg) and Mass B (1000 kg), are floating in space connected by a docking mechanism that acts as a spring with stiffness k. They are oscillating toward and away from each other. A student calculates the frequency using the formula f = (1/2pi) * sqrt(k/m) and uses m = 1000 kg. Why is this incorrect, and will the actual frequency be higher or lower than the student's calculation?

44) A critical flight computer is suspended by two identical springs to isolate it from launch vibrations. The springs are arranged in series (one attached to the other) to achieve a very soft suspension. If one of the springs jams and becomes rigid (effectively turning into a solid rod), what happens to the natural frequency of the system? (Calculate the factor of change).

45) A solar panel is mounted on the end of a deployable boom. The boom acts as a cantilever spring. The stiffness of a cantilever beam is k = (3 * E * I) / L^3, where L is the length. If the boom fails to deploy fully and is stuck at only half its intended length (L/2), but the mass of the solar panel remains the same, by what factor does the natural frequency of the vibration change?

46) An aerospace engineer is testing a prototype shock absorber for a lunar lander. The 500 kg lander is dropped from a small height onto the single leg, which contains a spring (k = 20,000 N/m) and a hydraulic damper with coefficient c.

a) If the engineer sets the damping coefficient to c = 8,000 Ns/m, will the lander oscillate after hitting the ground, or will it settle directly to equilibrium without crossing it? Identify the damping type (Underdamped, Overdamped, or Critically Damped).

b) What specific value of c (in Ns/m) would constitute "Critical Damping" for this system?

47) A loose panel on a cargo plane acts as a mass-spring-damper system with a natural frequency of 50 Hz and a damping ratio (zeta) of 0.05 (lightly damped). During flight, engine vibrations apply a driving force at a frequency of 50 Hz (resonance).

a) Calculate the Dynamic Amplification Factor (DAF) at this resonant frequency.

b) If the static displacement of the panel under a constant force of that magnitude is 0.5 mm, what is the actual amplitude of vibration during flight?

48) To ensure a wing doesn't flutter uncontrollably, engineers conduct a "pluck test." They displace the wingtip and release it. They measure the amplitude of the first peak to be 10 cm, and the amplitude of the second peak (one period later) to be 8 cm.

a) Calculate the logarithmic decrement (delta) for this motion.

b) Using delta, estimate the damping ratio (zeta) of the wing structure (assuming zeta is small). Is this system underdamped, overdamped, or critically damped?

49) A delicate satellite component is mounted on an isolation table to protect it from launch acoustics. The system has a natural frequency of 10 Hz and a damping ratio of zeta = 0.2. The launch noise creates a ground vibration at a frequency of 15 Hz.

a) Calculate the frequency ratio (r = driving frequency / natural frequency).

b) Calculate the Dynamic Amplification Factor (DAF) for this specific frequency ratio using the standard formula.

50) Liquid fuel sloshing inside a rocket tank is modeled as a damped mass-spring system. The effective mass of the fuel is m = 1000 kg. The stiffness is k = 90,000 N/m. The tank contains internal baffles that currently provide a damping coefficient of c = 15,000 Ns/m.

a) Construct the symbolic differential equation of motion for the displacement "x" of the fuel, plugging in the known numerical values for mass, damping, and stiffness.
b) Calculate the undamped natural frequency (omega_n) of this system in radians per second.
c) Calculate the critical damping coefficient (c_crit) and the damping ratio (zeta). Based on this ratio, is the system Underdamped, Overdamped, or Critically Damped?
d) The rocket's guidance computer adjusts the thrust vector at a frequency of 1.5 Hz. Convert your natural frequency from part (b) into Hertz. Is the system at risk of resonance?
e) Based on your answers to (c) and (d), is this specific configuration considered "Safe" or "Unsafe" for flight? Explain why, specifically referencing the combination of damping type and frequency match.
f) For a fuel control system, engineers usually aim for a damping ratio of zeta > 1. Why is this specific target physically desirable for liquid fuel?
g) Mathematically, the engineer needs to change the damping coefficient "c" to achieve zeta > 1. Physically, how could the engineer modify the tank internals to achieve this?

SOLUTIONS

1) 0.25 Hz
2) 8 m/s
3) 4.0 seconds
4) No through space, Yes through the hull.
5) It halves.
6) 1700 meters
7) It is at a Trough (Maximum Negative Displacement).
8) Parallel (Back and Forth); Neither.
9) Neither (Tie).
10) 5 meters

11) 0.1 seconds
12) Zero
13) Increases
14) 90 degrees
15) 3 meters
16) Increases
17) 3.0 meters
18) At Maximum Positive Displacement (Peak)
19) Frequency
20) 1 m/s

21) Answer: The signal will fade in and out.
22) Destructive Interference (Quiet)
23) Destructive Interference (Quiet)
24) Approximately 283 Hz
25) Intermediate Intensity
26) Weak (Destructive Interference)
27) One-quarter of a wavelength
28) Loud; It drops by 6 dB (to one-quarter intensity)
29) The Left transmitters must fire EARLIER
30) 1.0 meter

31) Equation: M * x_double_dot + (1.5k) * x = 0
Natural Frequency: sqrt(1.5k / M)
32) Equation: L * theta_double_dot + (4g/3) * theta = 0
Natural Frequency: sqrt(4g / 3L)
33) Equation: 5 * x_double_dot + 20 * x_dot + 500 * x = 0
Natural Frequency: 10 rad/s
34) Equation: (1.5M) * x_double_dot + k * x = 0
Natural Frequency: sqrt(k / 1.5M)
35) Equation: I * theta_double_dot + c * theta_dot + K_t * theta = 0
Required Damping c: 2 * sqrt(I * K_t)

36) 52.5 kg
37) a) Stays the same. b) Shifts by a factor of 5.
38) 4.0 Hz
39) The work done by gravity as the spring compresses (mg * x)
40) Approximately 789 m/s^2 (or about 80g)
41) Frequency is sqrt(k/m); Amplitude is "d"
42) No, it is unsafe (Resonance)
43) Incorrect mass used; The actual frequency will be HIGHER.
44) Frequency increases by a factor of approximately 1.41 (sqrt 2)
45) Increases by a factor of roughly 2.83

46) a) Overdamped (It will NOT oscillate).
b) approx 6,325 Ns/m.
47) a) DAF = 10.
b) Amplitude = 5.0 mm.
48) a) delta is approx 0.223.
b) zeta is approx 0.035 (3.5%); Underdamped.
49) a) Ratio r = 1.5.
b) DAF is approx 0.72 (or 72%).

50)
a) 1000 * x_double_dot + 15,000 * x_dot + 90,000 * x = 0
b) omega_n = square_root(90,000 / 1000) = square_root(90) = 9.49 rad/s.
c) Underdamped
d) Yes, there is a high risk of resonance
e) Unsafe. The system is at resonance (driving frequency matches natural frequency) AND it is underdamped
f) Ideally, the fuel should simply move to its new position without oscillating back and forth. Overdamping ensures the fuel returns to equilibrium sluggishly without crossing the center line, preventing the rhythmic weight shifting that confuses control systems.
g) To increase the damping coefficient "c", the engineer should add more baffles (or increase the surface area/roughness of existing baffles) inside the tank.